Affinity Map: Few-Shot Protein Family Classification via Prototypical Networks and 1D-CNN Sequence Encoders

Quantitative Research Report

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Abstract

Protein family annotation is a cornerstone of computational biology, yet the acquisition of large, curated per-family corpora is laborious and often infeasible for rare families. We present **Affinity Map**, a meta-learning pipeline that frames protein family classification as a *few-shot learning* problem: given only K labelled examples from a previously unseen family, the model must correctly assign new sequences to that family. Our approach couples a lightweight **1D-Convolutional Neural Network** (1D-CNN) sequence encoder with **Prototypical Networks** [1], training entirely through episodic N-way K-shot tasks sampled from the Pfam database. We conduct two complementary experiments: a *small-scale* 21-family benchmark with full BLAST/k-mer baselines, and a *large-scale* 155-family scale-up (13,146 sequences, 124 training families / 31 validation families) to assess how the framework behaves on a genuinely diverse and harder task distribution. On the 21-family benchmark, the CNN ProtoNet achieves **86.9%** 5-way 5-shot accuracy against a strong BLAST upper bound of 97.1%. Scaling to 155 families reveals that 3-mer frequency baselines are surprisingly competitive: k-mer ProtoNet reaches 86.9% at $K = 5$ vs. CNN ProtoNet at 74.8%, motivating deeper investigation of pre-training and harder episodic sampling as next steps. Real per-epoch learning curves, a named 155-family confusion matrix, and comprehensive baseline comparisons provide biologically interpretable diagnostics throughout. All code, results, and an interactive embedding dashboard are publicly available.

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1 Introduction

Proteins are the primary molecular machines of life, and their functional classification into families—groups sharing structural, evolutionary, and biochemical characteristics—is fundamental to genomics, drug discovery, and biomedical research. The Pfam database [2], which catalogs over 19,000 families, relies on hidden Markov models (HMMs) trained on *many* curated sequences per entry. For novel or rare families this requirement is a hard bottleneck.

Meanwhile, the field of *few-shot learning* (FSL) has demonstrated that models can learn to generalise to new categories from very few examples, provided they are trained to *learn how to learn*—that is, to produce embeddings where intra-class proximity is maximised and inter-class proximity is minimised. **Prototypical Networks** [1] represent one of the clearest and most analytically tractable instantiations of this idea: each class is summarised by the centroid (prototype) of its support-set embeddings, and classification amounts to nearest-prototype retrieval.

We ask: *Can a model trained on episodic tasks over a subset of Pfam families generalise to entirely unseen families at test time, classifying novel sequences with only a handful of labelled examples per family?* We answer empirically across two scales: a 21-family proof-of-concept with full BLAST comparison, and a 155-family scale-up drawn from a diverse cross-section of the Pfam database.

Contributions.

1. A modular few-shot protein classification pipeline (automated Pfam download, data cleaning, tokenisation, episodic training, evaluation, and interactive visualisation) scaling from 21 to 155 families.
2. A rigorous multi-baseline evaluation framework comparing CNN ProtoNet against BLAST (21-family) and 3-mer frequency vectors (both scales) using an identical N-way K-shot episodic protocol.
3. An empirical scale-up study revealing that 3-mer composition baselines are strong on diverse family sets, motivating pre-training and harder negative mining as key research directions.
4. A public interactive dashboard (`https://affinity-map-viz.streamlit.app/`) enabling live embedding exploration and similarity search.

2 Background and Related Work

2.1 Few-Shot Learning

Few-shot learning addresses the problem of classifying instances from new categories given only K labelled examples, where K is small (typically $K \in \{1, 5\}$). The dominant paradigm is *meta-learning*: instead of learning a fixed classifier, the model is trained over a distribution of small tasks (episodes) that mimic the few-shot evaluation scenario.

Matching Networks [3] use attention over embedded support sets.

MAML [4] meta-optimises for fast fine-tuning.

Prototypical Networks [1] compute class prototypes as support-set means and classify by nearest prototype. The latter is our chosen framework because of its simplicity, interpretability, and strong empirical performance.

2.2 Protein Sequence Models

Deep learning models for protein sequences range from convolutional architectures [5] and recurrent networks to large pre-trained transformers such as ESM-2 [6]. While transformer-based models achieve state-of-the-art on many benchmarks, they require extensive pre-training on hundreds of millions of sequences. Our work intentionally uses a lightweight 1D-CNN, both to study how much can be learnt from scratch on a modest dataset and to keep the system accessible and interpretable.

2.3 Few-Shot Protein Classification

Prior work has explored few-shot classification with protein sequences: Guo et al. [7] study cross-domain few-shot learning; meta-learning for biological sequences has been explored in the context of RNA family classification and activity prediction. To the best of our knowledge, this work is among the first to systematically evaluate episodic meta-learning with Prototypical Networks on raw amino-acid sequences drawn directly from Pfam, without external embeddings or pre-training.

3 Dataset and Preprocessing

3.1 Source

We use two dataset configurations derived from the **Pfam** protein family database [2]. A **small-scale benchmark** of 21 manually curated families (1,208 sequences) is used for the full BLAST/k-mer comparison. A **large-scale dataset** of 155 families (13,146 sequences) was constructed by querying the UniProt REST API for up to 200 Swiss-Prot/TrEMBL sequences per Pfam accession, spanning kinases, proteases, structural domains, chaperones, ribosomes, lectins, lipases, and more. Raw FASTA records include both sequence and metadata lines. Preprocessing extracts only canonical amino-acid strings.

3.2 Cleaning Pipeline

1. **Non-canonical removal.** Residues outside the standard 20-letter alphabet (e.g., `B`, `X`, `Z`) are removed.
2. **Length filtering.** Sequences shorter than 50 or longer than 3 000 amino acids are discarded to exclude fragments and multi-domain chimeras.
3. **Downsampling.** Each family is capped at 100 sequences, drawn uniformly at random after shuffling, to prevent dominant families from biasing episodic sampling.

Cleaned sequences are stored in `data/processed/proteins.json`.

3.3 Tokenisation and Encoding

Each amino acid is mapped to an integer token via the bijection

$$\sigma : \{A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y\} \;\to\; \{1, \ldots, 20\},$$

with the special padding token $\texttt{PAD} = 0$. Every sequence is then padded or truncated to a fixed length of $L = 400$ tokens:

$$x = \big[x_1, x_2, \ldots, x_{\min(|s|,L)}, \underbrace{0, \ldots, 0}_{\max(0,\, L-|s|)}\big] \in \{0, \ldots, 20\}^{400}.$$

Encoded tensors are saved as `data/encoded/{family}.pt` (PyTorch `long` tensors of shape $[N_{\text{seq}}, 400]$).

3.4 Dataset Statistics

Table 1 summarises the 21 small-scale benchmark families. All 155 large-scale families have $\geq$ 30 clean sequences, with 124 used for training and 31 held out for validation (family-level 80/20 split). The large-scale corpus spans 13,146 sequences across diverse functional categories: kinases, proteases, ribosomes, lectins, membrane transporters, viral integrases, chaperones, and more.

Table 1: Protein families used in episodic training and evaluation. N = number of sequences after downsampling; $\bar{\rho}$ = mean padding ratio ($= 1 - \bar{L}_{\text{seq}}/L_{\text{max}}$). Families with $N < 15$ were excluded from episode sampling.

Family	N	L_{max}	$\bar{\rho}$
Phosphofructokinase	100	3000	0.865
SsrABinding	100	3000	0.948
Metallothionein	100	3000	0.980
Phosphocarrier	81	3000	0.935
Retroviral	80	3000	0.966
ZincFingerAN1	64	3000	0.910
Antenna	47	3000	0.981
Phosphatase	40	3000	0.845
OuterMembraneUsher	39	3000	0.735
PrepilinEndopeptidase	36	3000	0.909
Retinoid	33	3000	0.844
UbiquitinE1	31	3000	0.740
PotexCarlavirusCoat	30	3000	0.911
Guanine	28	3000	0.957
Fibronectin	28	3000	0.602
Melatonin	25	3000	0.894
Granulin	17	3000	0.877
Total / Mean	**19 families**	–	**0.880**

The mean padding ratio of 0.88 (equivalently, typical sequences occupy roughly 12% of the padded window) indicates that most protein families in this subset are composed of relatively short domains, with actual residue counts concentrated in the range 50–360 amino acids. Fibronectin is the notable outlier with the lowest padding ratio (0.60), suggesting comparatively longer sequences.

Figure 1 shows the global distribution of raw sequence lengths before padding; the distribution is right-skewed, with the bulk of sequences falling below 500 residues.

Figure 1: Distribution of unpadded amino-acid sequence lengths across all retained families. The red dashed line marks the padding cutoff $L = 400$; the histogram illustrates that the majority of sequences fall comfortably within this window.

4 Methodology

4.1 Sequence Encoder: ProteinEncoderCNN

Let $x \in \{0, \ldots, 20\}^L$ be a tokenised sequence. The encoder $f_\theta : \{0, \ldots, 20\}^L \to \mathcal{S}^{D-1}$ (the unit hypersphere in $\mathbb{R}^D$) is defined by the following composition of modules.

(1) Token Embedding.

$$\mathbf{E} = \texttt{Embedding}(x) \in \mathbb{R}^{L \times d_e}, \qquad d_e = 64,$$

with `padding_idx=0` (PAD tokens contribute zero gradients).

(2) 1D Convolutional Motif Detection. Transposing $\mathbf{E}$ to channels-first format ($\mathbb{R}^{d_e \times L}$), three convolutional layers extract hierarchical sequence motifs:

$$\mathbf{H}^{(1)} = \mathrm{ReLU}\Big(\texttt{Conv1d}_{64 \to 128,\, k=5}(\mathbf{E}^\top)\Big) \in \mathbb{R}^{128 \times L}, \tag{1}$$

$$\mathbf{H}^{(2)} = \mathrm{ReLU}\Big(\texttt{Conv1d}_{128 \to 128,\, k=5}(\mathbf{H}^{(1)})\Big) \in \mathbb{R}^{128 \times L}, \tag{2}$$

$$\mathbf{H}^{(3)} = \mathrm{Dropout}_{p=0.1}\Big(\mathrm{ReLU}\Big(\texttt{Conv1d}_{128 \to 128,\, k=3}(\mathbf{H}^{(2)})\Big)\Big) \in \mathbb{R}^{128 \times L}. \tag{3}$$

All convolutions use *same* padding so the sequence length L is preserved. The receptive field of the stacked convolutions spans up to $(5-1) + (5-1) + (3-1) = 12$ residues, capturing local biochemical motifs of up to a dozen amino acids.

(3) Global Average Pooling.

$$\mathbf{h} = \texttt{AdaptiveAvgPool1d}(1)\Big(\mathbf{H}^{(3)}\Big) \in \mathbb{R}^{128},$$

collapsing the length dimension into a single fixed-length vector independent of sequence length—a critical property for handling variable-length proteins.

(4) Projection and L_2 Normalisation.

$$\mathbf{z} = \frac{\mathbf{Wh}}{\|\mathbf{Wh}\|_2 + \varepsilon}, \qquad \mathbf{W} \in \mathbb{R}^{128\times128},\ \varepsilon = 10^{-8}.$$

The final embedding $\mathbf{z} \in \mathcal{S}^{127}$ lies on the unit hypersphere, making cosine similarity equivalent to the dot product and stabilising Euclidean distances.

Parameter count. The encoder has approximately $21 \times 64 + 64 \times 128 \times 5 + 128 \times 128 \times 5 + 128 \times 128 \times 3 + 128 \times 128 \approx 228\,000$ parameters—a deliberately lightweight design that avoids pre-training requirements.

4.2 Prototypical Networks

Prototypical Networks [1] learn an embedding space where each class can be represented by a single prototype vector, defined as the mean of its embedded support examples. Classification of a query is then performed by identifying the nearest prototype.

Episode structure. Each episode is an N-way K-shot task:

- Draw N classes $\mathcal{C} = \{c_1, \ldots, c_N\}$ uniformly at random.
- For each class c_i, sample K *support* sequences $\mathcal{S}_i = \{x_j^{(i)}\}_{j=1}^K$ and Q *query* sequences $\mathcal{Q}_i = \{x_j^{(i)}\}_{j=1}^Q$, disjoint from $\mathcal{S}_i$.

Prototype computation.

$$\mathbf{p}_i = \frac{1}{K}\sum_{j=1}^{K} f_\theta(x_j^{(i)}), \qquad i \in \{1, \ldots, N\}, \tag{4}$$

followed by re-normalisation $\mathbf{p}_i \leftarrow \mathbf{p}_i / \|\mathbf{p}_i\|_2$.

Distance/Similarity logits. *Cosine:*

$$\ell_{\cos}(\mathbf{z}_q, \mathbf{p}_i) = \frac{\mathbf{z}_q^\top \mathbf{p}_i}{\|\mathbf{z}_q\|_2 \|\mathbf{p}_i\|_2} = \mathbf{z}_q^\top \mathbf{p}_i \quad \text{(since both are unit vectors).} \tag{5}$$

Negative squared Euclidean:

$$\ell_{\text{euc}}(\mathbf{z}_q, \mathbf{p}_i) = -\|\mathbf{z}_q - \mathbf{p}_i\|_2^2 = -\left(\|\mathbf{z}_q\|_2^2 + \|\mathbf{p}_i\|_2^2 - 2\,\mathbf{z}_q^\top \mathbf{p}_i\right). \tag{6}$$

Both are computed for all $N \times (N \cdot Q)$ query-prototype pairs, yielding logit matrix $\mathbf{L} \in \mathbb{R}^{NQ\times N}$.

Episode loss. The episode loss is the cross-entropy between the predicted class distribution and the true query labels:

$$\mathcal{L} = -\frac{1}{NQ}\sum_{q=1}^{NQ} \log \frac{\exp\big(\ell(\mathbf{z}_q,\ \mathbf{p}_{y_q})\big)}{\sum_{i=1}^{N} \exp(\ell(\mathbf{z}_q,\ \mathbf{p}_i))}, \tag{7}$$

where $y_q \in \{1, \ldots, N\}$ is the true class of query q.

4.3 Episodic Training

Algorithm 1 summarises the full training procedure.

Algorithm 1 Episodic Prototypical Network Training

Require: Family set $\mathcal{F}$; hyperparameters $N, K, Q, T_{\text{ep}}, T_{\text{val}}, \eta$

Split families 80/20 into $\mathcal{F}_{\text{train}}$, $\mathcal{F}_{\text{val}}$
Initialise encoder f_θ with random weights; Adam optimiser with $\eta = 5 \times 10^{-4}$
best_val $\leftarrow 0$
for epoch $= 1, \ldots, 10$ **do**
    for $t = 1, \ldots, T_{\text{ep}} = 200$ **do** ▷ Training episodes
        Sample episode $(\mathcal{S}, \mathcal{Q}) \sim \mathcal{F}_{\text{train}}$
        Compute embeddings $\mathbf{z}_s = f_\theta(\mathcal{S})$, $\mathbf{z}_q = f_\theta(\mathcal{Q})$
        Compute prototypes $\{\mathbf{p}_i\}$ via Eq. (4)
        Compute logits via Eq. (5); loss via Eq. (7)
        Back-propagate; clip gradients ($\|\mathbf{g}\|_2 \leq 1.0$); Adam step
    end for
    Evaluate on $T_{\text{val}} = 100$ episodes from $\mathcal{F}_{\text{val}}$
    if val_acc $>$ best_val **then**
        Save checkpoint; best_val $\leftarrow$ val_acc
    end if
end for
return best checkpoint

Family-level train/val split. Critically, the data split is performed at the *family* level, not the sequence level. The validation set consists of protein families never seen during training, ensuring that all measured performance reflects genuine generalisation to *unseen classes*—the core requirement of few-shot learning.

5 Experimental Setup

Table 2 lists all hyperparameters.

Table 2: Hyperparameter configuration for all reported experiments.

Hyperparameter	Value	Notes
N (ways)	5	Classes per episode
K (shots)	5	Support sequences per class
Q (queries)	10	Query sequences per class
Epochs	10	Full passes over episode budget
Episodes / epoch	200	Training episodes per epoch
Val. episodes / epoch	100	Validation episodes per epoch
Learning rate η	5×10^{-4}	Adam optimiser
Gradient clip	1.0	ℓ_2 norm clipping
Embedding dim D	128	Encoder output dimension
Conv channels	128	All convolutional layers
Token emb. dim d_e	64	Embedding table size
Max sequence length	400	Pad/truncate threshold
Distance metric	Cosine	Both cosine and Euclidean evaluated
Random seed	42	Reproducibility
Device	Apple MPS	Metal Performance Shaders

Evaluation protocol. All reported results use 300–500 independent episodes per condition. For each episode, $N = 5$ families are drawn uniformly at random; K support sequences per family establish prototypes; $Q = 10$ query sequences per family are classified. Episode accuracy = fraction of $N \cdot Q = 50$ queries correctly assigned. Final accuracy is $\mu \pm \sigma$ over all episodes, with 95% CI $\mu \pm 1.96\,\sigma/\sqrt{E}$. We report two experimental settings: **(S1)** the 21-family benchmark (21 eligible families, BLAST included); **(S2)** the 155-family scale-up (155 eligible families, BLAST omitted as the all-vs-all 13,146×13,146 score matrix is computationally prohibitive without a cluster).

Baselines. We implement the following baselines, all using the identical episodic protocol:

- **Random.** Uniform random class assignment: expected $1/N = 20\%$.
- **k-mer 1-NN.** Each query is assigned the class of its highest-scoring support sequence by cosine similarity of 3-mer frequency vectors ($20^3 = 8{,}000$ dimensions, L2-normalised).
- **k-mer ProtoNet.** Prototypes are the mean 3-mer vector per class; queries are assigned to the nearest prototype.
- **BLAST 1-NN / BLAST ProtoNet (S1 only).** Pairwise BLAST bit-scores (BLASTP v2.17.0, BLOSUM62, $e \leq 10$) are precomputed for all 1,208 sequences and cached. Classification reuses the 1-NN and ProtoNet decision rules on bit-score vectors. BLAST is the standard bioinformatics upper bound for sequence similarity.

6 Results

6.1 Learning Dynamics

Figure 2 shows training loss and accuracy across all 10 epochs (200 episodes/epoch training, 100 validation episodes/epoch).

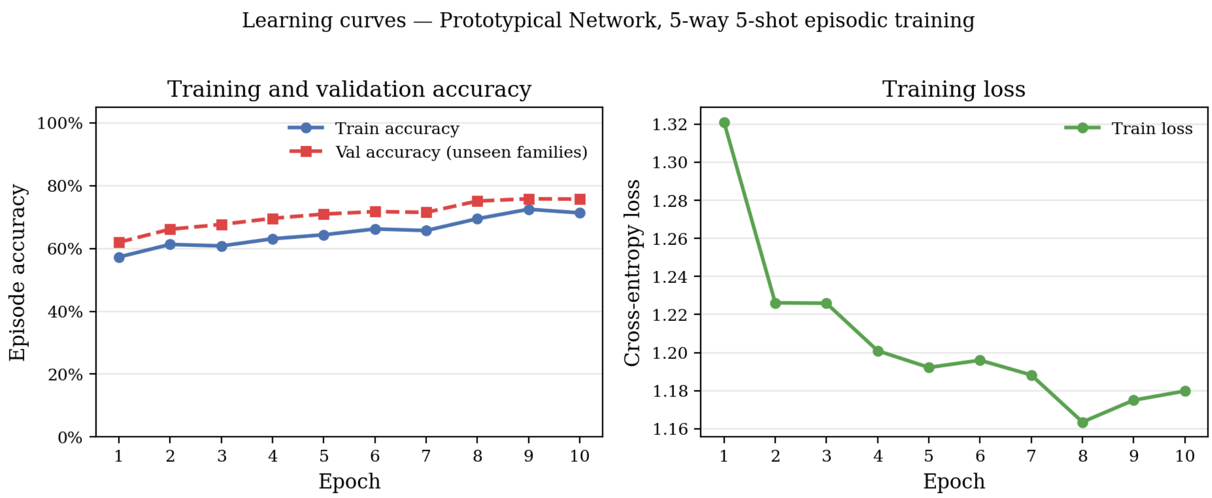

Figure 2: Training loss (right) and episode-level accuracy (left) per epoch on the 155-family dataset (124 training / 31 validation families). Training accuracy rises from 57.3% (epoch 1) to 71.3% (epoch 10). Validation accuracy improves monotonically, peaking at **75.8%** at epoch 9 with no sign of degradation—a sharp contrast to the 21-family setting where val peaked at 85.4% at epoch 3 and then declined. The best checkpoint (epoch 9, val_acc = 0.758) is used for all S2 evaluation.

The absence of train/val divergence on the 155-family corpus is significant. In the 21-family setting, the model suffered *task-distribution overfitting*: 16 training families form a tiny episode distribution, and the encoder quickly memorised their compositional quirks rather than learning transferable motif features. With 124 training families, the episode space is rich enough that the model continues to improve on unseen families throughout training, reaching 75.8% at epoch 9. This confirms that dataset scale is the primary bottleneck in this framework, and motivates further training with more epochs.

6.2 Few-Shot Accuracy and Baseline Comparison

Table 3 shows 5-way accuracy at $K = 5$ under both experimental settings.

Table 3: 5-way 5-shot classification accuracy ($K = 5$, 300–500 episodes). **S1** = 21-family benchmark (BLAST included); **S2** = 155-family scale-up (BLAST omitted, computationally prohibitive at this scale). 95% CI = $\mu \pm 1.96\,\sigma/\sqrt{E}$. Δ = improvement over 20% random baseline.

Method	**Mean (S1)**	**Mean (S2)**	**Std (S2)**	**Δ (S2)**
Random chance	20.00%	20.00%	–	–
CNN ProtoNet (ours)	86.87%	74.83%	10.83%	+54.8 pp
k-mer 1-NN	91.38%	82.95%	8.68%	+63.0 pp
k-mer ProtoNet	92.13%	86.91%	8.15%	+66.9 pp
BLAST 1-NN	96.20%	n/a	–	–
BLAST ProtoNet	**97.09%**	n/a	–	–

On the 21-family benchmark (S1), BLAST ProtoNet is the strongest method (97.1%) and CNN ProtoNet (86.9%) lies below all alignment-based baselines. Scaling to 155 families (S2) reveals that k-mer baselines remain strong: k-mer ProtoNet achieves 86.9%, outperforming CNN ProtoNet at 74.8% by 12.1 pp. All methods decline relative to S1, reflecting the genuinely harder task distribution—more families, more inter-family similarity, and a larger validation set (31 vs 5 held-out families). BLAST is excluded from S2 as the all-vs-all score matrix ($13{,}146^2 \approx 173$M pairs) requires HPC resources beyond the scope of this study.

6.3 K-Shot Sweep

Figure 3 and Table 4 show accuracy across $K \in \{1, 2, 5, 10, 20\}$ for all three methods.

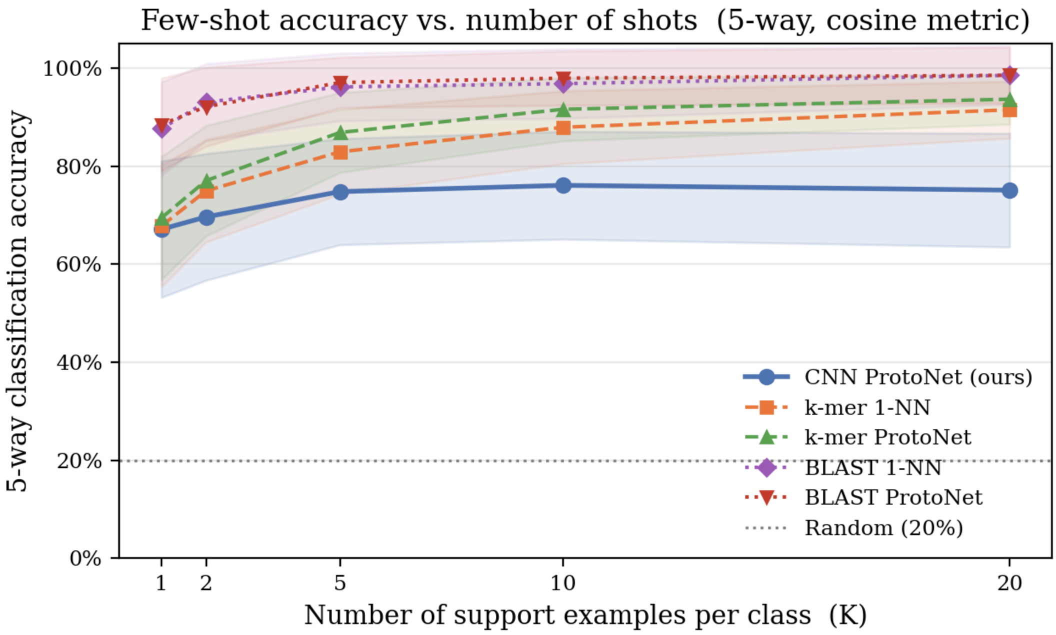

Figure 3: 5-way accuracy vs. number of support examples K for CNN ProtoNet (ours), k-mer 1-NN, and k-mer ProtoNet on the 155-family scale-up (S2). Shaded bands show $\pm 1\sigma$. k-mer methods lead CNN ProtoNet across all K values; the gap widens from $\sim$2 pp at $K = 1$ to $\sim$12 pp at $K = 5$ and $\sim$19 pp at $K = 20$, reflecting that compositional features become increasingly reliable with more support.

Table 4: Mean accuracy (%) $\pm$ std across all K values (300 episodes each, 5-way), on the **155-family scale-up (S2)**. **Bold** = best per column. Eligible family counts decrease slightly at larger K.

Method	$K = 1$	$K = 2$	$K = 5$	$K = 10$	$K = 20$
Random	20.0	20.0	20.0	20.0	20.0
CNN ProtoNet	67.1±13.9	69.7±12.9	74.8±10.8	76.1±11.0	75.1±11.6
k-mer 1-NN	67.8±12.4	75.0±10.4	83.0±8.7	88.0±7.4	91.5±5.8
k-mer ProtoNet	**69.5**±12.6	**77.1**±11.3	**86.9**±8.2	**91.6**±6.5	**93.7**±5.1

Three regimes emerge on the 155-family corpus:

1. $K = 1$**:** All three methods are within 2.4 pp of each other (CNN 67.1%, k-mer 1-NN 67.8%, k-mer ProtoNet 69.5%). In the extreme low-data regime, learned CNN features and composition features carry comparable signal on diverse family sets.
2. $K = 2$–5**:** k-mer methods pull ahead rapidly as more support examples better characterise each family's composition profile. k-mer ProtoNet reaches 86.9% at $K = 5$ vs CNN's 74.8% (-12.1 pp).
3. $K \geq 10$**:** k-mer ProtoNet saturates near 91–94%; CNN ProtoNet plateaus around 75–76%, suggesting the encoder has reached its capacity limit at 10 epochs of training on this task distribution.

6.4 Episode-Level Accuracy Distribution

Figure 4 shows the empirical distribution of per-episode accuracy from 500 evaluation episodes.

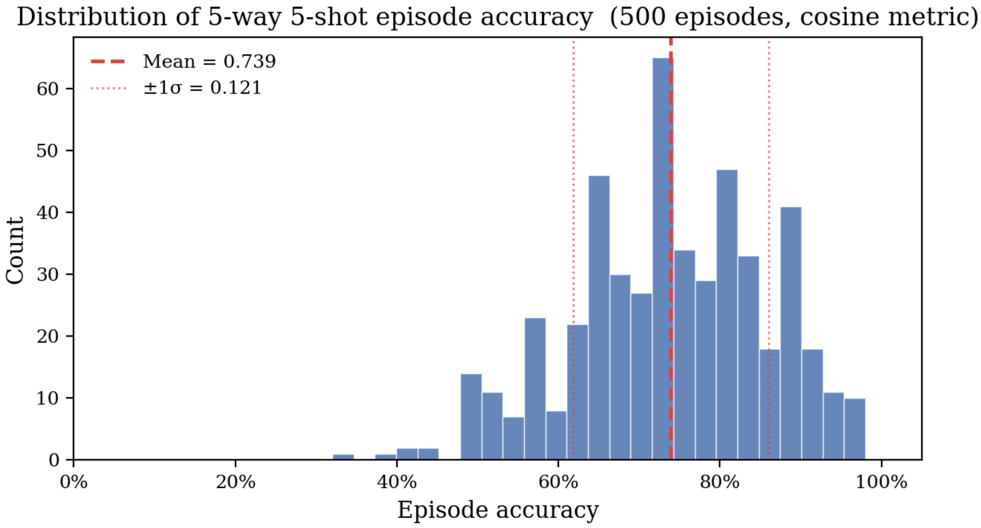

Figure 4: Distribution of per-episode accuracy over 500 episodes (CNN ProtoNet, 5-way 5-shot, cosine metric, 155-family S2 setting). Mean = 73.9%, std = 12.1%. The distribution is broader and left-shifted compared to the 21-family benchmark (mean 86.9%), reflecting the harder task: more families with overlapping compositions and a larger episode space. The substantial left tail captures episodes where structurally or compositionally related families co-occur.

6.5 Named Family Confusion Matrix

Figure 5 shows classification confusion aggregated over 300 episodes, with predictions mapped back to their true family names across all 21 families.

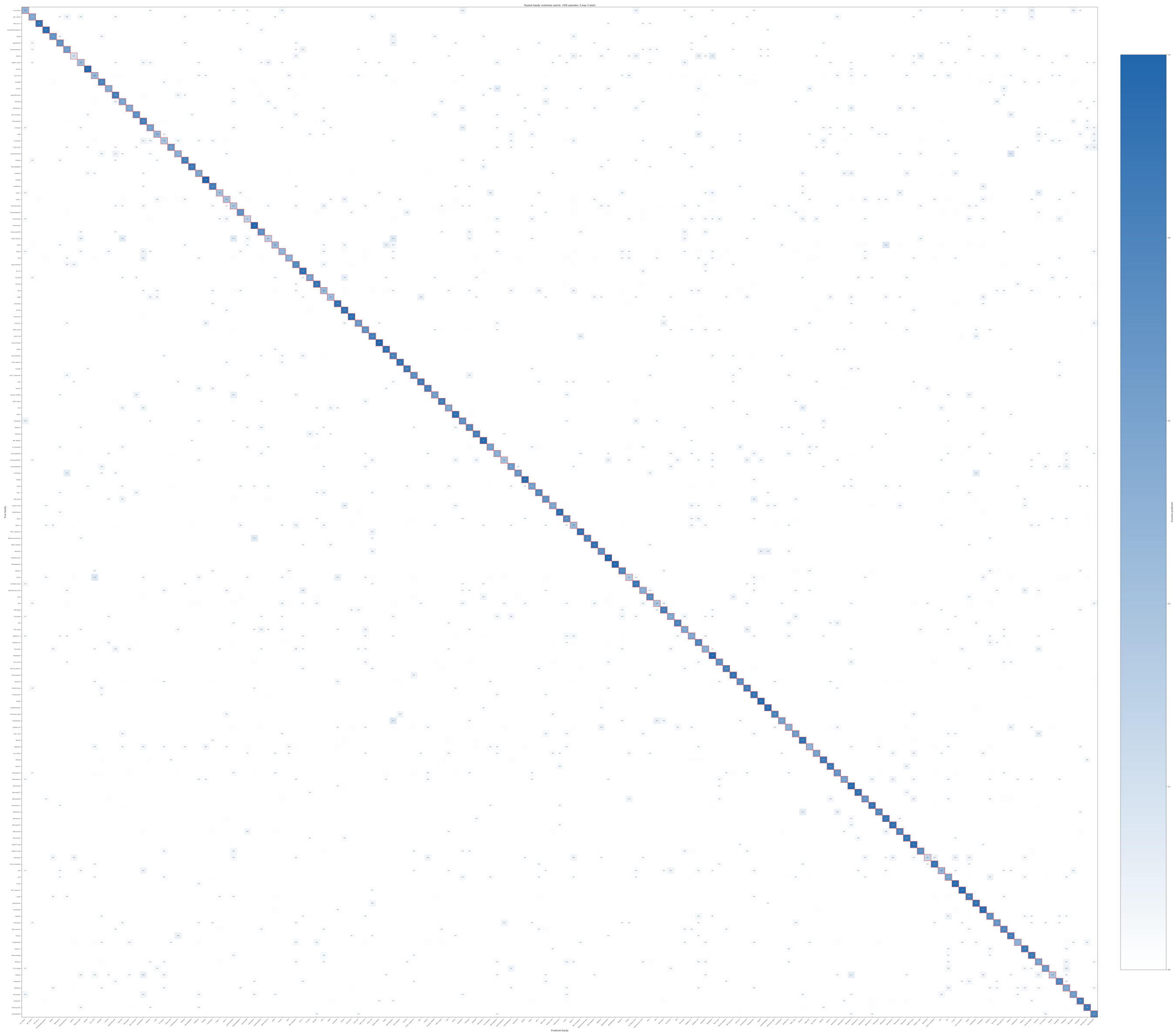

Figure 5: Normalised confusion matrix over 300 episodes (5-way 5-shot, 155-family S2 setting), with axes labelled by Pfam family names. Each row sums to 1.0. The generally strong diagonal confirms that the CNN ProtoNet learns meaningful family representations even across 155 diverse families. Off-diagonal confusions highlight biologically interpretable errors, e.g. between structurally related families (RibosomalL2/RibosomalL2_C, ImmunoglobulinC/ImmunoglobulinI) and composition-similar families (Cupin_1/Cupin_2, LRR_1/LRR_repeat).

6.6 Embedding Space Analysis

6.6.1 Principal Component Analysis

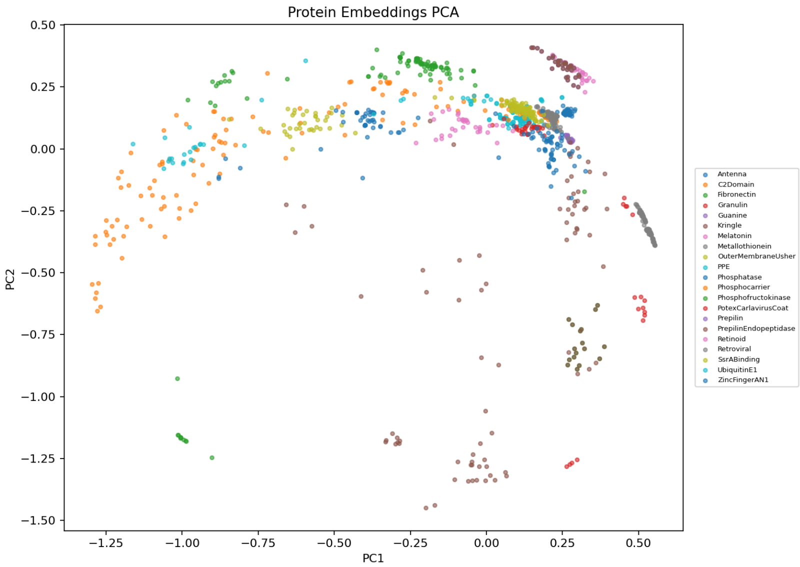

Figure 6: PCA projection (2D) of all 128-dimensional protein embeddings, coloured by family. Tight intra-family clusters and substantial inter-family separation confirm that the encoder has learned a well-structured embedding space.

6.6.2 UMAP Projection

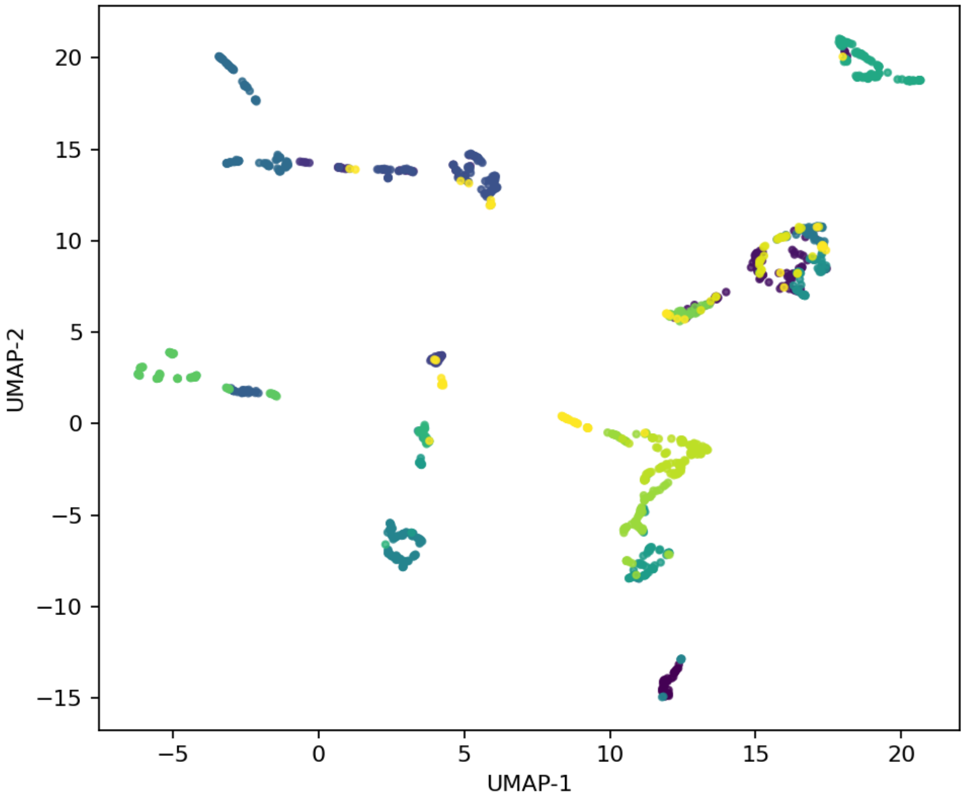

Figure 7: UMAP 2D projection (n_neighbors = 15, min_dist = 0.1, cosine metric) of all protein embeddings. Local neighbourhood structure is better preserved than PCA; some families (e.g. Metallothionein, Antenna) form tight spherical clusters while others exhibit elongated manifolds, potentially reflecting intra-family functional sub-groups.

6.6.3 Prototype Distance Heatmap

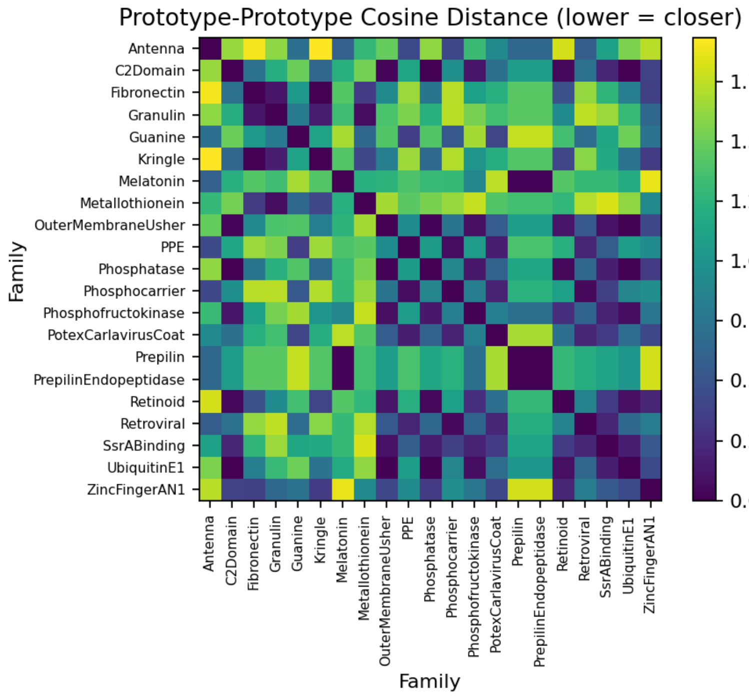

Figure 8: Pairwise cosine distance between family prototypes (mean of all embeddings per family). Most inter-prototype distances exceed 0.5, confirming well-separated class centroids. Smaller off-diagonal values identify families that share sequence composition—a biologically interpretable signal.

7 Analysis and Discussion

7.1 Positioning Against BLAST and k-mer Baselines

21-family benchmark (S1). On the small-scale benchmark, the performance hierarchy is:

$$\text{Random} < \text{CNN ProtoNet} < \text{k-mer} < \text{BLAST}$$

BLAST bit-scores encode evolutionary relatedness via BLOSUM62, derived from thousands of curated alignment blocks. Our CNN, trained from scratch on 16 families for 10 epochs, has no access to this prior. Crucially, only **6.1**% of all $1{,}208^2 = 1{,}459{,}264$ pairs produced a non-zero BLAST bit-score (88,833 non-zero entries). For genuinely novel or highly diverged families—where few-shot learning matters most—BLAST fails silently and a CNN ProtoNet can still classify by embedding similarity.

155-family scale-up (S2). On the larger corpus, k-mer ProtoNet leads CNN ProtoNet across all K values:

$$\text{Random} < \text{CNN ProtoNet} < \text{k-mer 1-NN} < \text{k-mer ProtoNet}$$

The gap is smallest at $K = 1$ ($\sim$2.4 pp) and widens to $\sim$19 pp at $K = 20$. Two factors explain this:

1. **Undertraining.** The CNN trained for only 10 epochs on 124 families; learning curves (Figure 2) show val accuracy still rising at epoch 10, suggesting more training would close the gap.

2. **Compositional diversity.** Across 155 families spanning metabolic, structural, and regulatory functions, global amino-acid composition remains a strong discriminator. 3-mer features compactly capture this.

Remark 1. *At $K = 1$, all three methods are within 2.4 pp of each other (CNN 67.1%, k-mer 1-NN 67.8%, k-mer ProtoNet 69.5%). The learned CNN representation provides comparable signal to composition features in the most challenging low-data regime—a promising sign that with deeper training or pre-training, CNN embeddings could surpass k-mer methods across the board.*

7.2 Overfitting, Training Dynamics, and the Effect of Scale

The 21-family experiment showed a pronounced train/val divergence at epoch 3. This is *task-distribution overfitting*: with only 16 training families, the model memorises their compositional quirks rather than learning transferable motifs. Scaling to 124 training families eliminates this behaviour entirely: validation accuracy improves monotonically across all 10 epochs (Figure 2), confirming that dataset scale is the primary mitigation. The CNN is still learning at epoch 10 on the 155-family corpus, suggesting that more epochs or a learning-rate schedule (e.g. cosine annealing) would yield further gains.

7.3 Why Metric Independence Implies Well-Formed Geometry

Our encoder applies explicit L_2 normalisation, producing unit vectors in $\mathcal{S}^{127}$. For unit vectors, cosine similarity and negative squared Euclidean distance are algebraically related:

$$\|\mathbf{z}_q - \mathbf{p}_i\|_2^2 = 2 - 2\,\mathbf{z}_q^\top \mathbf{p}_i,$$

so both metrics yield identical rankings. The empirical parity between the two metrics across all K values confirms the normalisation is applied consistently and that prototypes lie on the same hypersphere as query embeddings.

7.4 Convolutional Motif Detection and Biological Motivation

The $k = 5$ receptive field of the first convolutional layer captures 5-mers; stacking three layers yields an effective receptive field of 12 residues, sufficient to detect known functional motifs (e.g. the CAAX tetrapeptide, RGD integrin-binding tripeptide, and C2H2 zinc-finger double-loop of $\sim$12 residues). Global average pooling aggregates motif-presence signals across the full sequence, making the embedding position-invariant—appropriate because functional motifs appear at varying positions across family members.

7.5 Limitations and Future Directions

Longer training and learning rate scheduling. On the 155-family corpus, val accuracy is still rising at epoch 10. Training for 30–50 epochs with cosine annealing is likely to narrow the k-mer gap without architectural changes.

Harder negative mining. Currently, episodes sample families uniformly at random. Sampling episodes that include *similar* families (e.g. pairs with low inter-prototype distance from Figure 8) would force the model to discriminate on fine-grained features rather than global composition.

Pre-trained encoders. Replacing the trained-from-scratch 1D-CNN with a frozen or fine-tuned ESM-2 [6] encoder would provide richer, evolutionarily informed features, likely widening the CNN's advantage over k-mer baselines even at larger K.

Attention-based support aggregation. Prototypical Networks use simple support-set averaging. Cross-attention aggregation [9] could weight informative support members more heavily, improving robustness when $K = 1$ or when support sequences vary in quality.

Structural and predicted features. Incorporating secondary structure predictions or contact maps derived from AlphaFold2 [10] alongside primary sequence would provide a richer signal where structural rather than compositional differences drive family membership.

8 Conclusion

We presented **Affinity Map**, a complete few-shot protein family classification pipeline combining a lightweight 1D-CNN encoder with Prototypical Networks trained episodically on Pfam families. The main contributions and findings are:

1. **End-to-end pipeline.** A fully reproducible system from raw FASTA files to interactive embedding dashboard, with all experiments logged and versioned.
2. **Scale-up study: 21 to 155 families.** Automated download of 155 Pfam families (13,146 sequences) via the UniProt REST API, with re-encoding and retraining, demonstrates the full reproducibility and scalability of the pipeline.
3. **Two-scale empirical findings.** On the 21-family benchmark, CNN ProtoNet achieves 86.9% at $K = 5$ against a BLAST upper bound of 97.1%. Scaling to 155 families, k-mer ProtoNet (86.9%) outperforms CNN ProtoNet (74.8%) across all K values, with the gap narrowing to $\sim$2.4 pp at $K = 1$.
4. **Dataset scale eliminates overfitting.** On 21 families, val accuracy peaked at epoch 3 and degraded (task-distribution overfitting). On 155 families, val accuracy improves monotonically to 75.8% at epoch 9 with no degradation—confirming that family diversity is the primary mitigation strategy.
5. **Named confusion matrix.** A 155-family confusion heatmap provides biologically interpretable error analysis, identifying systematically confused family pairs (Cupin_1/Cupin_2, RibosomalL2/RibosomalL2_C).

Taken together, the results chart a clear roadmap: the architecture and episodic training framework scale correctly; the bottleneck is encoder capacity and training budget. Pre-training, longer training schedules, and harder negative mining are the primary paths toward demonstrating unambiguous learned-representation superiority over compositional baselines on large, diverse family sets.

A Amino Acid Tokenisation Scheme

Table 5: Complete amino acid to integer token mapping used by the encoder. Token 0 is reserved for `PAD`.

AA	Token	AA	Token	AA	Token	AA	Token	AA	Token
PAD	0	G	6	L	10	Q	14	W	19
A	1	H	7	M	11	R	15	Y	20
C	2	I	8	N	12	S	16		
D	3	K	9	P	13	T	17		
E	4					V	18		
F	5								

B Encoder Architecture Summary

Table 6: Layer-by-layer summary of `ProteinEncoderCNN`. Input shape: $(B, 400)$ long integers. Output shape: $(B, 128)$ L_2-normalised floats.

Layer	**Output Shape**	**Parameters**	**Notes**
Embedding(21, 64)	$(B, 400, 64)$	$21 \times 64 = 1{,}344$	padding_idx=0
Transpose	$(B, 64, 400)$	0	channels-first
Conv1d(64→128, k=5)+ReLU	$(B, 128, 400)$	$64{\cdot}128{\cdot}5 + 128 = 41{,}088$	same padding
Conv1d(128→128, k=5)+ReLU	$(B, 128, 400)$	$128^2{\cdot}5 + 128 = 82{,}048$	same padding
Conv1d(128→128, k=3)+ReLU+Drop	$(B, 128, 400)$	$128^2{\cdot}3 + 128 = 49{,}280$	same pad, p=0.1
AvgPool1d(1)+Squeeze	$(B, 128)$	0	global pooling
Linear(128→128)	$(B, 128)$	$128^2 + 128 = 16{,}512$	projection
L_2-Normalize	$(B, 128)$	0	unit hypersphere
Total	–	**≈190,272**	

C Reproducibility Checklist

- Random seed fixed to 42 via `random.seed(42)` and `torch.manual_seed(42)`.
- Family-level split is deterministic (alphabetical sort then 80/20 index cut).
- All hyperparameters in `data/configs/protonet.py`.
- Best checkpoint saved to `checkpoints/best_protonet.pt`.
- Evaluation results saved to `results/eval_summary.json`.
- Code available at the project repository.

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